

The reserve limit

$R = -\varphi_{bb}$ prefers a floor $R_\infty \approx 0.03$; the log model is killed

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August 29, 2026

Abstract

Round 420 left a shrinking reserve $R = -\varphi_{bb} = |c_J| - \Sigma$ along selected $a_k = 2^k$ (0.105, 0.068, 0.052, 0.048, 0.038) and asked whether $R \rightarrow 0$ or $R \rightarrow R_\infty > 0$. That is the switch for the cofinal edge. Five selected points plus EXT prefer a floor $R_\infty \approx 0.030$ (AIC); $R \sim c/\log k$ is killed; $k = 10$ border-fails. No RH claim.

Status. Lemma 1 is **reduced**. $R = -\varphi_{bb}$ and the T_0 - $C_{\min}$ linearisation are finite identities (Propositions 1, 2). The log model is refuted as a selected law (Proposition 3). Ausgang FLOOR_PREFERRED / LOG_KILLED / K10_BORDER_FAIL / COFINAL_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (reserve limit). *On the selected sequence $a_k = 2^k$, $R = -\varphi_{bb} = |c_J| - \Sigma$ converges to a strictly positive floor $R_\infty > 0$ (equivalently: the edge does not saturate $R = 0$).*

Verdict: REDUZIERT. A five-point AIC census on $k = 5, \dots, 9$ prefers $R_\infty \approx 0.030$ versus k (0.034 versus $a_k = 2^k$). $M3$ ($R \sim c/\log k$) is killed ($\Delta\text{AIC} > 15$). $M2$ ($R \rightarrow 0$ as a power) loses by $\Delta\text{AIC} \approx 7$. EXT vacuous windows sit in $[0.029, 0.037]$. $k = 10$ ($k_z = 197$, $N = 4071$) border-fails; $k = 11$ ($N = 12508$) is skipped. The 0-versus-0.03 split is therefore not seen at the hoped depth. Not a theorem.

2 Identities

Proposition 1 (reserve). $R = -c_J - \Sigma = -\varphi_{bb}$. Over $\mathbb{Q}$ (vacuous toy) $c_J = -\frac{3}{2}$, $\Sigma = \frac{1}{2}$ gives $R = 1 = -\varphi_{bb}$. At $k_z = 9$, $R = 0.08562$.

Proposition 2 (T_0 dictionary). By the r411 identity $\mathfrak{T}_0^* \mathfrak{T}_0 = C^{-1}$, $\|\mathfrak{T}_0\| = C_{\min}^{-1/2}$. If $C_{\min} = 1 + \varepsilon$ then

$$\|\mathfrak{T}_0\| - 1 = -\frac{1}{2}\varepsilon + O(\varepsilon^2). \quad (1)$$

On vacuous $k_z = 26, 43$ the ratio $(\|\mathfrak{T}_0\| - 1)/(-(C_{\min} - 1)/2)$ equals 1.0000. This is the razor scale, of size 10^{-6} - 10^{-7} on those windows.

3 The measurement

Selected $a_k = 2^k$ (live $k = 4, 5, 6, 7, 9$; $k = 8$ pinned from calibration, wall 226 s, builder fallback; N is not monotone):

k	k_z	N	chart	R	den	Σ
4	9	184	P1	0.08562	1.601	0.313
5	17	96	P1	0.10493	1.651	0.244
6	26	364	VAC	0.06811	1.524	0.408
7	43	839	VAC	0.05198	1.518	0.430
8	69	5690	P1	0.04763	1.569	0.384
9	116	1433	P1	0.03814	1.482	0.480

Diagnostic models on $k = 5, \dots, 9$ versus the selected index k (not versus N):

	model	R_∞	AIC	rms
M1	$R_\infty + c k^{-\alpha}$	+0.0298	-57.4	0.0018
M2	$c k^{-\alpha}$	0	-50.4	0.0043
M3	$c / \log k$	0	-38.9	0.0167

Proposition 3 (AIC). *M1 wins. M3 is killed ($\Delta\text{AIC} > 15$). M2 loses ($\Delta\text{AIC} \approx 7$). Versus $a_k = 2^k$ the same order holds ($R_\infty = +0.034$). Versus N the abscissa is illegal ($k = 8$ has $N = 5690 > k = 9$'s 1433); that fit is discarded. Dropping $k = 9$ does not flip the winner.*

EXT vacuum $R \in [0.0285, 0.0373]$ sits in the M1 floor band.

4 What drives R

den is flat in $[1.48, 1.65]$ (gap-to-2 is $O(0.5)$, not shrinking). Σ wanders in $[0.24, 0.48]$ and is the balance partner. Three Σ -channels, measured separately:

- $\|s\|^2 \sim 0.06$ is flat — not the driver.
- mass-weighted λ falls mildly $5.5 \rightarrow 3.8$, then bounces ($k = 8$: 4.52) — not a collapse onto the razor.
- $n_{\text{eff}} = 1 / \sum p_i^2$ grows $18 \rightarrow 146$ ($k = 5..9$) and 559 at $k = 8$: occupation *broadens*.

The razor share of Σ stays $\leq 3 \cdot 10^{-7}$. $B_w = S_{N-2} + 5/7$ grows $6.4 \rightarrow 18.9$ on the selected range: the border-mass series $S = \sum \rho_k$ is not convergent here. $\gamma = \|b\|^2 / B_w$ falls only slowly ($0.71 \rightarrow 0.58$).

So Σ moves because the s -occupation gets wider, not because $\|s\|^2$ grows and not because the occupied λ crash into 1.

5 Five saturation numbers, three scales

quantity	selected VAC	rate / size	dictionary
$2 - \text{den}$	0.48	flat $O(0.5)$	budget; <i>not</i> $C_{\min}$
$C_{\min} - 1$	$10^{-6} - 10^{-7}$	razor, falling	r407
$\ \mathfrak{T}_0\ - 1$	$-10^{-6} - 10^{-7}$	$= \frac{1}{2}(1 - C_{\min})$	r411, (1)
$ \text{sch} $	0.055–0.069	tracks R	r417: $\text{sch} = \varphi_{bb} - \tau^2$
R	0.052–0.068	floor ≈ 0.03 or $\alpha \approx 0.34$ vs 2^k	r418/r420

The r407/r411 pair converts $C_{\min} - 1$ into $\|\mathfrak{T}_0\| - 1$. The r417/r418 pair converts sch into R once $\tau \rightarrow 0$. The budget gap $2 - \text{den}$ is a third object and stays $O(1)$. These are *not* one saturation scale and they do not share one fundamental exponent. The edge-balance scale — R itself — is the quantity the lemma is about; its selected law is the M1 floor, not a shared razor rate.

6 Kills

Dead vacuous χ : $R = -0.036, -0.027, -0.054$, all negative, no trend toward rescue. Overflow six: all $R < 0$ (they live by τ at finite N , r419). Fit mutant (drop $k = 9$): $M1$ still wins with $R_\infty > 0.02$. Scramble $\text{nneg}(A_0) = 21$ (bulk dead). Drop-border: no reserve scalar.

7 What is a theorem, what is a census

Propositions 1 and 2 are theorems (finite identities). Proposition 3 is a named refutation of $M3$ plus an AIC census for $M1$. Lemma 1 is not a theorem: $R_\infty > 0$ is preferred, not proved; $k = 10/11$ are missing; $R \rightarrow 0$ remains a losing but living alternative on a wrong abscissa. If the floor holds, the cofinal edge reduces to showing $\Sigma \rightarrow \Sigma_\infty < |c_{J,\infty}|$ with $\text{den}_\infty < 2$ already $O(1)$. If $R \rightarrow 0$, the edge saturates and the cofinal argument has the same depth as P1. The measurement leans toward the first reading and does not close either. No RH claim.